FINAL YEAR PROJECT PROPOSAL

Umer Hussain

SalesChamp

AI-Powered Multi-Agent Sales Automation Platform with Salesforce CRM, Microsoft Dynamics 365 Sales, and SAP ERP Integration

Abstract

In today's competitive business landscape, sales teams spend a significant portion of their time performing repetitive administrative tasks such as lead discovery, customer research, proposal creation, CRM updates, and ERP coordination. These manual activities reduce productivity, delay customer engagement, and increase operational costs.

SalesChamp aims to transform the traditional sales workflow by introducing an AI-powered multi-agent platform capable of automating the complete sales lifecycle from identifying potential customers to managing quotations, generating proposals, updating CRM records, synchronizing ERP transactions, and assisting sales representatives throughout the customer journey.

Unlike conventional CRM systems that primarily act as repositories of customer information, SalesChamp functions as an intelligent sales assistant capable of reasoning, planning, and executing complex workflows using specialized AI agents. **By integrating Salesforce CRM, Microsoft Dynamics 365 Sales, and SAP ERP, the platform bridges customer relationship management and enterprise resource planning, enabling seamless synchronization between customer interactions, sales activities, and enterprise business operations across multiple platforms.**

The proposed system leverages Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, cloud computing, Microsoft Copilot technologies, and enterprise APIs to automate business processes while maintaining human oversight for critical decision-making.

Problem Statement

Sales professionals often spend more time on administrative work than actual selling. Typical challenges include:

- Manual lead generation
- Researching prospective customers
- Creating personalized proposals
- Updating CRM systems

- Coordinating with finance and inventory departments
- Managing quotations and invoices
- Following up with customers
- Synchronizing business information across multiple enterprise platforms

Existing CRM systems require users to perform these tasks manually or through predefined workflows, limiting scalability and efficiency.
Furthermore, CRM platforms rarely integrate intelligent decision-making capabilities with ERP systems, causing information silos and delayed business operations.

Proposed Solution

The platform integrates:

- Salesforce CRM
- Microsoft Dynamics 365 Sales
- SAP ERP
- AI Agents
- Retrieval-Augmented Generation (RAG)
- Voice AI
- Business Intelligence Dashboard
- Microsoft Power Platform (Future Integration)

Instead of acting as a chatbot, SalesChamp functions as an autonomous Sales Development Representative (SDR) capable of making informed decisions, interacting with enterprise systems, and assisting human sales teams.

Objectives

The primary objectives of SalesChamp are:

- Automate lead generation using web search and business intelligence.
- Perform AI-powered customer research.
- Generate personalized sales proposals.
- Automatically update Salesforce CRM.
- Integrate SAP ERP for quotations, invoices, inventory, and order processing.
- Implement intelligent follow-up through email and AI voice calls.
- Provide predictive lead scoring using machine learning.
- Offer executive dashboards with sales analytics.
- Demonstrate practical implementation of Agentic AI within enterprise software.
- Automatically synchronize customer data between Salesforce and Microsoft Dynamics 365 Sales.
- Enable AI-powered CRM interoperability across multiple enterprise CRM platforms.
- Utilize Microsoft Dynamics 365 Sales for opportunity management, forecasting, and AI Copilot-assisted selling.

Key Features

AI Lead Discovery

The platform automatically discovers businesses through:

- Google Maps
- Business Directories
- Company Websites
- LinkedIn (where permitted by APIs and terms)
- Public business databases

The AI identifies companies with high sales potential.

Intelligent Business Research

Multiple research agents analyze:

- Company profile
- Products
- Competitors
- Digital presence
- Technology stack
- Estimated business size
- Pain points
- Industry trends

The collected information is stored in a centralized knowledge base.

AI Proposal Generator

Using RAG and LLMs, the platform generates:

- Website development proposals
- Software proposals
- AI implementation proposals
- Cloud migration proposals
- Digital transformation proposals

Each proposal is personalized according to the prospect's business requirements.

Salesforce CRM Integration

SalesChamp automatically:

- Creates Leads
- Creates Contacts
- Updates Opportunities
- Logs customer interactions
- Tracks communication history
- Manages sales pipeline
- Predicts conversion probability

Sales representatives can continue working directly from Salesforce while AI handles routine updates.

SAP ERP Integration
Once a prospect becomes a customer, SalesChamp synchronizes data with SAP.
Examples include:

- Customer Master creation
- Quotations
- Sales Orders
- Inventory verification
- Product availability
- Invoice generation
- Delivery scheduling
- Payment status

This eliminates duplicate data entry between CRM and ERP systems.

CRM Agent
The CRM Agent manages customer information across multiple CRM platforms.
Responsibilities include:

- Create Leads in Salesforce
- Create Leads in Dynamics 365
- Update Opportunities
- Synchronize Contacts
- Maintain Customer Activities
- Forecast Sales Pipeline
- Remove duplicate customer records
- Synchronize CRM data

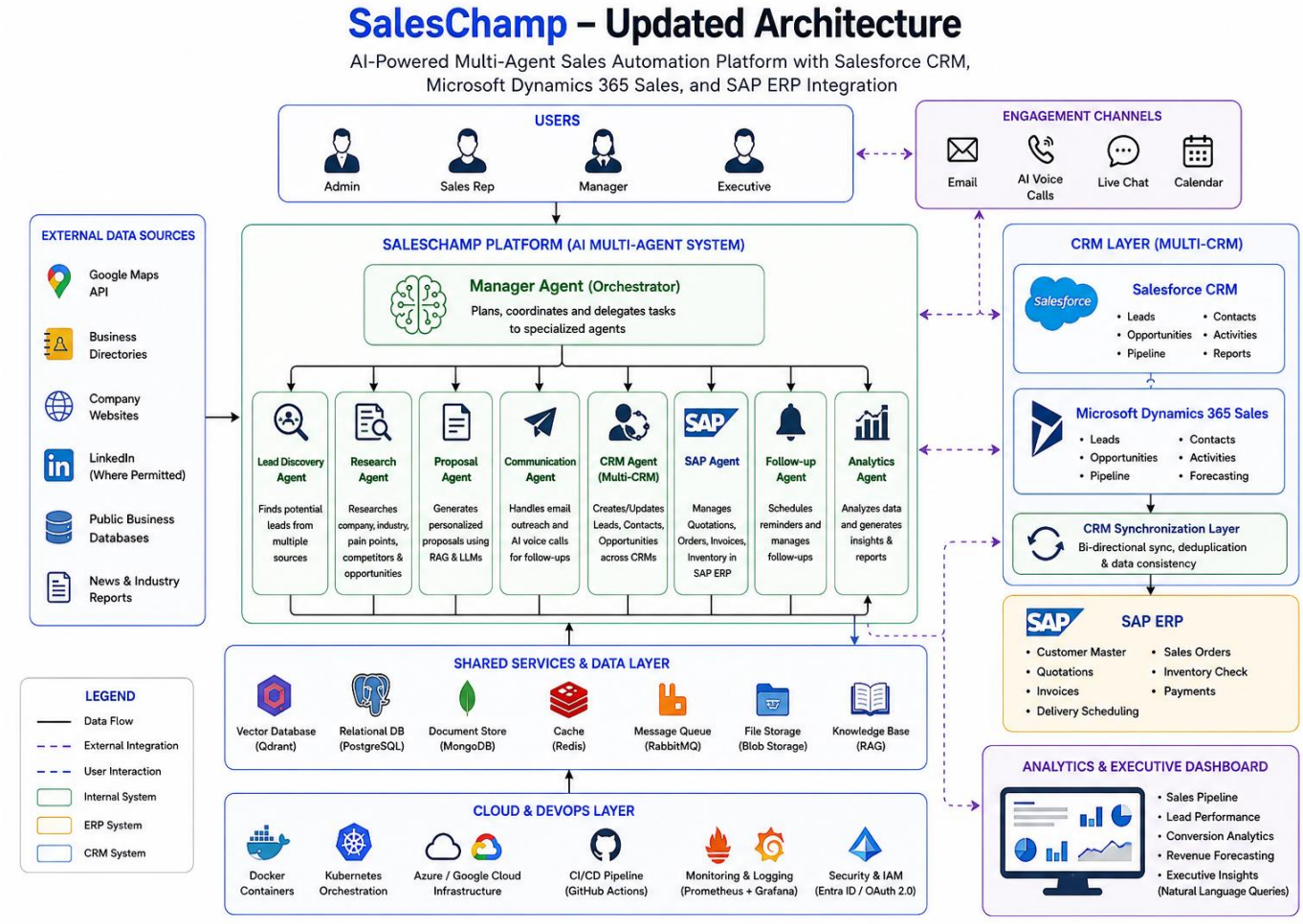
Microsoft Dynamics 365 Sales Integration
SalesChamp provides seamless integration with **Microsoft Dynamics 365 Sales**, enabling organizations using Microsoft's ecosystem to benefit from AI-powered automation.
The platform automatically:

- Create Leads
- Create Contacts
- Create Accounts
- Create Opportunities
- Update Sales Pipelines
- Record Customer Activities
- Schedule Meetings
- Generate Sales Forecasts

- Synchronize Customer Information
- Trigger Power Automate Workflows
- Support Microsoft Copilot recommendations

By supporting both Salesforce and Microsoft Dynamics 365 Sales, SalesChamp becomes CRM-independent and can adapt to different enterprise environments without changing the AI workflow.

Proposed System Architecture



Proposed Technology Stack

Frontend

- React.js / Next.js
- Tailwind CSS

Backend

- Django REST Framework or FastAPI

AI Framework

- LangGraph
- Google ADK
- OpenAI Agents SDK

LLMs

- GPT-5
- Gemini
- Azure OpenAI

Vector Database

- Qdrant
- ChromaDB

Database

- PostgreSQL

CRM

- Salesforce REST APIs
- Salesforce Apex APIs
- Microsoft Dynamics 365 Web API
- Microsoft Dataverse
- Microsoft Power Platform
- Microsoft Graph APIERP
- SAP BTP APIs
- SAP OData Services

Authentication

- OAuth 2.0
- JWT

Cloud

- Microsoft Azure
- Google Cloud

DevOps

- Docker
- Kubernetes

AI Workflow

Find Lead ↓ Research Company ↓ Analyze Pain Points ↓ Generate Proposal ↓ Human Approval ↓ Email / Voice Outreach ↓ Create Lead ↓ Choose CRM (Salesforce OR Microsoft Dynamics 365 Sales) ↓ CRM Synchronization ↓ Lead Qualification ↓ Opportunity Management ↓ SAP ERP ↓ Quotation ↓ Inventory Check ↓ Sales Order ↓ Invoice ↓ Analytics Dashboard

Innovative Features

- Multi-Agent AI Architecture
- Salesforce CRM Integration
- SAP ERP Integration
- Retrieval-Augmented Generation
- Autonomous Proposal Generation
- AI Voice Outreach
- Human-in-the-loop Approval
- Predictive Lead Scoring
- Executive Dashboard
- Real-time Analytics
- Multi-CRM Architecture
- Microsoft Dynamics 365 Sales Integration
- CRM Synchronization Engine
- Microsoft Copilot Ready
- Cross-Platform Enterprise Automation

Expected Outcomes

SalesChamp is expected to:

- Reduce manual sales effort
- Improve lead conversion rates
- Increase sales productivity
- Minimize CRM data entry
- Automate ERP synchronization
- Improve customer response time
- Provide intelligent business insights
- Demonstrate enterprise AI automation using Agentic AI
- Eliminate duplicate customer records across Salesforce and Dynamics 365.
- Enable seamless interoperability between Microsoft and Salesforce ecosystems.
- Improve enterprise sales collaboration through synchronized CRM and ERP data.

Future Scope

Potential future enhancements include:

- Microsoft Dynamics 365 Customer Service
- Microsoft Dynamics 365 Customer Insights
- Microsoft Dynamics 365 Field Service
- Microsoft Copilot Studio Agents
- Microsoft Fabric Analytics
- Power BI Executive Dashboards
- SAP S/4HANA Integration

- Oracle ERP Integration
- HubSpot Integration
- WhatsApp Business AutomationHubSpot integration
- WhatsApp Business automation
- Multi-language support
- AI negotiation agent
- Contract generation
- E-signature integration
- Mobile application
- AI sales coaching
- Predictive revenue forecasting

Novelty of the Project

The key innovation of SalesChamp lies in its ability to unify **Agentic AI**, **multi-CRM integration**, and **ERP automation** within a single intelligent platform. Unlike existing CRM automation solutions that operate within a single ecosystem, SalesChamp provides seamless interoperability between **Salesforce CRM**, **Microsoft Dynamics 365 Sales**, and **SAP ERP**, allowing AI agents to manage customer interactions, synchronize enterprise data, and automate business processes across heterogeneous enterprise systems.

Through a **Multi-Agent AI Architecture**, Retrieval-Augmented Generation (RAG), and enterprise API integrations, the platform autonomously discovers leads, conducts business research, generates personalized proposals, manages sales opportunities, coordinates CRM activities, synchronizes ERP transactions, and provides executive analytics while maintaining human oversight for high-impact decisions.

This combination of **Agentic AI, multi-CRM orchestration, ERP integration, and cloud-native enterprise architecture** distinguishes SalesChamp from traditional CRM systems and positions it as a modern, scalable solution for intelligent digital sales transformation.